Smart Collectorate

Taking a District Magistrate's office from paper files to an AI-enabled workflow, and getting people to use it

Pulse Rajan · Applied AI Engineer assessment · Creditfix Group · November 2026

The problem: district backlogs are growing

1 crore+

grievances on CPGRAMS in the last five years

157 → 15

days to dispose, central ministries, 2014 → 2025

82.1%

closed within the 21-day target in 2025, so about 1 in 5 is late

2.16 lakh

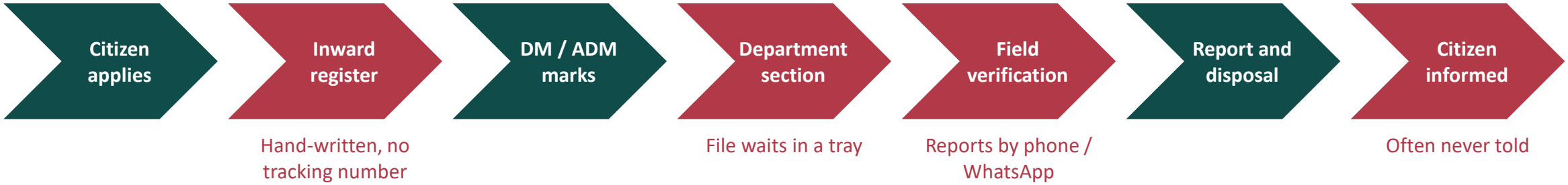
pending with States/UTs (June 2026), up ~26% since Dec 2025

In a typical district office (our working assumptions, measured in discovery)

- ~4,000 grievances and applications a month, ~30 days to dispose, 1 in 3 citizens come back only to ask for status
- Paper files and registers, follow-ups on phone and WhatsApp, no single owner per step, no live view for the DM

Sources: DARPG CPGRAMS monthly reports (PIB, 2026); Lok Sabha replies as reported by ThePrint and All India Radio News (2025–26); data.gov.in.

A. Discovery: map the workflow with its people



Who we talk to (25–30 interviews)

- DM, ADM, SDMs: which three numbers do you want every Monday?
- Clerks and inward desk: walk me through one file
- Field staff: how do reports reach you and go back?
- Citizens at the gate: how many visits did this take?

Assumptions we check, and how

- Mostly paper, mostly Hindi → count one month of the register
- ~10 categories cover 70% → hand-tag 300 past cases
- Data stays in India → confirm with NIC
- CPGRAMS / IGRS stays the system of record → check API access

2–3 weeks · trace 50 real files with dates at every hand-off · rank steps by delay × volume

A. 50 questions, each to the person who can answer it

Who we ask	Department / office	Qs	Example question
DM / ADM	Collector's office	1–6	Q4. How are joint problems (sewer + water + power) handled today, and who leads?
Grievance clerk	Inward desk	7–12	Q10. How often does a file come back as “not our department”?
Tehsildar / SDM, Lekhpal	Revenue	13–19	Q14. How do you ask another department for a report today, and how long does it take?
Executive Engineers, EO	Electricity, Jal Nigam, Municipal, PWD	20–29	Q24. When supply fails because of a sewer, power cut or road digging, who coordinates?
DSO, BDO, DSWO	Food, Rural Development, Welfare	30–35	Q30. Which ration complaints can be checked in the ePoS data without a visit?
CMO, SP grievance cell	Health, Police	36–39	Q39. What police information must never be shown to other departments?
District Informatics Officer	NIC	40–45	Q42. May masked text go to an AI service, or must AI run in India?
Citizens	Gate and helpline	46–50	Q47. Did you have to go to more than one office for one problem?

One interview per role, 25–30 in total · each department gets its own questions · full list of 50 in the appendix

A. Research: where the time really goes

Slow step	Why it is slow	Typical delay*	What fixes it
Inward entry and marking	Hand-written diary; file waits for DM / ADM to mark it	**2–5 days**	Number at entry; AI suggests the department
Sent to the wrong department	Clerk guesses; file bounces back and forth	**+5–10 days**	AI routing; transfer only with a reason
Reports from other departments	Letter, then reminders; nobody owns the follow-up	**7–15 days**	Tracked report request with a deadline
Field verification	Patwari / JE visit; report by paper or WhatsApp	**7–20 days**	Mobile app: photo, location, deadline
Approval and signature queue	File sits on one desk; officer on leave or transferred	**3–7 days**	Queue visible to DM; delegate on leave
Status enquiries	Citizens visit or call; staff stop work to search	**~1 in 3 cases**	SMS / WhatsApp status by tracking ID
Weekly review preparation	Clerks count figures by hand from registers	**1–2 days a week**	Live dashboard, report in one click
Closed but not fixed	Closed on paper; citizen complains again	**~9% reopened**	Citizen confirms before closing; spot checks

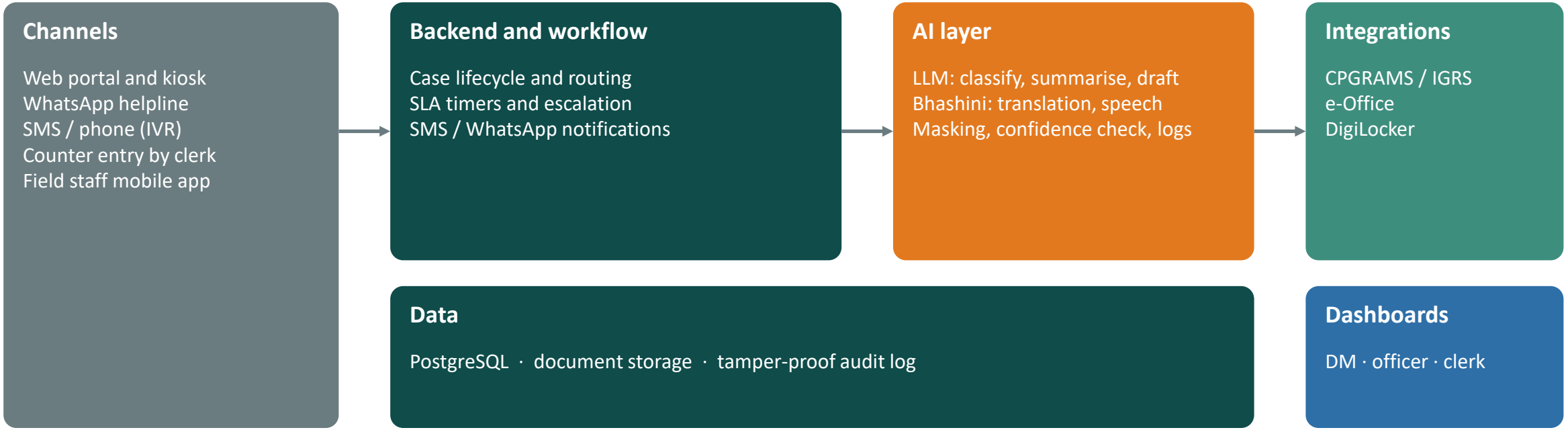
**Working estimates from published audits and news reports, replaced by real timings from 50 traced files in discovery. Reopen rate ~9%: Andhra Pradesh grievance audit (The Hans India). Department silos and mechanical closures: Jalandhar (Indian Masterminds), Markapuram (Deccan Chronicle). Delhi now needs HoD approval to transfer a grievance (Daily Pioneer).*

B. Eight use cases, and where AI is really needed

#	Use case	Why it matters	AI or rules?	Phase
1	Grievance intake, splitting and routing	Right department(s) on day one; splits complaints with several problems	AI	Pilot
2	QR code tracking on every file	No lost files; anyone can see who holds a file now	Rules	Pilot
3	SLA reminders and escalation	Files stop waiting in trays unnoticed	Rules	Pilot
4	Live dashboard for the DM	Backlog and overdue cases by department, today	Rules	Pilot
5	Inter-department requests and joint cases	Transfers, report requests and shared cases tracked, not on WhatsApp	Rules	Pilot
6	WhatsApp helpline (Hindi / English)	File and check status without visiting	AI	Rollout
7	Weekly review-meeting reports	Minutes instead of two days of compiling	Rules + AI	Rollout
8	AI assistant for officers	Summarise files, draft replies, find past orders	AI	Scale

Our rule: AI only where the input is messy language. Tracking, deadlines, alerts and dashboards are plain rules: cheaper, predictable, auditable. AI suggests; a person decides.

Architecture, and one grievance start to finish



1. Citizen files

Any channel, Hindi or English

2. Gets a number

Tracking ID by SMS

3. AI suggests

Unsure? A clerk confirms

4. Officer acts

Sees summary and deadline

5. Reminders

At 50% and 90%, then ADM

6. Closed

Citizen told and rates it

B. One complaint, three problems: where does it go?

> **“Meri gali mein sewer overflow ho raha hai, isliye paani nahi aa raha, aur bijli bhi baar-baar ja rahi hai.”**

My street’s sewer is overflowing, so water isn’t coming, and the power keeps going off.

Citizen gets ONE number: GRV-1042

AI splits

GRV-1042-A — **Sewer overflow → Municipal body / Sanitation**

LEAD · root cause · health risk → High · 3 days

GRV-1042-B — **No water supply → Jal Nigam / PHED**

Linked to A: water team sees it is waiting on sewer · 5 days

GRV-1042-C — **Power cuts → Electricity (DISCOM)**

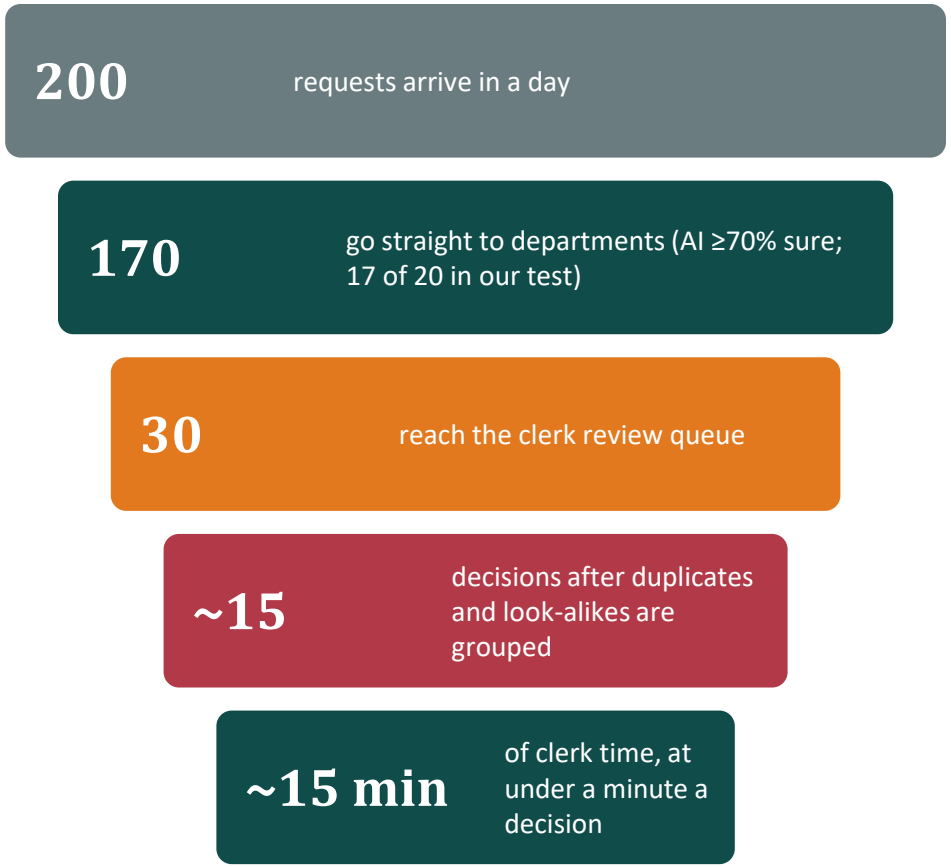
Independent: starts at once · 2 days

The rules

- AI lists each separate problem; each becomes a child ticket with its own department, officer and deadline
- Linked problems get a lead department (the root cause); the others can see the link, so no one is blamed for waiting
- Unsure split or under 70% confidence → a clerk checks before anything is sent
- The SDM / ADM owns the joint case: sees all parts on one screen, calls a joint site visit if needed
- Citizen gets an SMS as each part is fixed; GRV-1042 closes only when all three are closed
- The dashboard counts multi-department cases separately, so the DM sees where departments must work together

B. What if 200 requests land on one clerk?

The clerk is not the router. AI routes most cases straight to departments; the clerk only sees the unclear ones, and the queue never blocks a citizen.



Urgent first, not first-come

Queue sorted by urgency and deadline; health and safety cases on top.

Group and act in bulk

40 complaints about one transformer become one group, routed with one click; duplicates merged.

One-click confirm

AI shows its top two departments and a summary; the clerk confirms, no re-reading the whole text.

Share the load

Queues per tehsil, auto-shared between clerks with a daily cap; backlog over 50 alerts the nodal officer.

Never stuck

Not reviewed in 24 hours? It goes to the AI's best department, flagged "please confirm".

Fewer each month

Every correction teaches the system; better intake form. Target: review share falls from ~15% to under 10%.

Illustrative day, based on our 20-case test (85% auto-routed); real rates measured in the pilot.

B. How departments work together inside the system

Today: letters, phone calls and WhatsApp groups. Nothing is linked to the case, so nobody can see who is holding it up.

Transfer with a reason

Wrong department? One click, reason required. The receiver accepts within 1 day or it goes to the ADM. A second transfer needs ADM approval, so no ping-pong.

Report requests

Revenue asks PWD for a site report or NOC with a deadline. Tracked like a task, with reminders and escalation.

Case notes, not WhatsApp

Messages live inside the case and are logged. A new officer sees the full history on day one.

Joint cases

One lead department and supporting ones, a shared timeline, and an SDM / ADM who owns the whole case.

Circulars from the DM office

Orders and circulars to all departments, with read receipts and acknowledgement.

Alerts that reach people

In-app, SMS and WhatsApp alerts, a daily digest for officers, and e-Office for formal noting.

Every hand-off is timed, so the dashboard shows who each case is waiting on. All six are working in the live prototype.

C. Roadmap: zero to district-wide in 12 months

Discovery

Weeks 1–3

- Workflow map, baseline
- Use cases signed off

Team: Lead, AI engineer, UX researcher

~8 person-weeks

Pilot

Weeks 4–14

- 2 departments, 1 tehsil
- Use cases 1–5 live
- 4 weeks parallel with paper

Team: Lead, 2 devs, AI engineer, QA, trainer

~45 person-weeks

Rollout

Months 4–6

- Whole district
- WhatsApp helpline
- Weekly reports

Team: Pilot team + 1 dev, 2 trainers

~70 person-weeks

Scale

Months 7–12

- Officer AI assistant
- Field app, integrations

Team: 2 devs, AI engineer, QA, DevOps

~80 person-weeks

Support

Ongoing

- Helpdesk and fixes
- Monthly AI review

Team: Support engineer, helpdesk

~1.5 people

Go / no-go at week 14: we scale only if the pilot targets are met. Office side: one nodal officer (~20% time) and a champion clerk per department.

What a successful pilot looks like

−40%

average time to dispose of a case

−30%

backlog older than 30 days

≥ 85%

AI routing accuracy (confirmed by officers)

≥ 80%

of officers using it every week

≥ 4 / 5

citizen rating of the service

0

lost files in pilot departments

All measured against the discovery baseline, for the same departments, after 10 weeks live.

D. If we hold citizen data: how it stays safe

The Collectorate stays the owner (Data Fiduciary). We are only the processor, under a signed data processing agreement.

Law and contract

- DPDP Act 2023 and DPDP Rules 2025: notice in Hindi / English, purpose limits, citizen rights
- IT Act and CERT-In directions: report incidents within 6 hours, keep logs 180 days
- Aadhaar never stored in full: masked, or a vault reference only
- Data stays in India: NIC, State Data Centre or MeitY-empanelled cloud

Technical controls

- Encrypted in transit and at rest; phone and Aadhaar encrypted field by field
- Encryption keys held by the Collectorate, not by us
- Role and area based access, OTP login; tamper-proof log of every view and change
- Masked before AI; no citizen data used to train models; India-hosted AI if required
- Encrypted daily backups and a recovery site

Process and people

- Collect only what is needed; delete or anonymise on a fixed schedule
- CERT-In empanelled audit and penetration test before go-live, then yearly
- Staff background checks and NDAs; access reviewed every quarter
- Breach plan: tell the office at once, CERT-In in 6 hours, the Data Protection Board and citizens as the Rules require
- Exit: full data export and certified deletion

No one on our team can open a citizen record in production without a logged, approved request.

D. Risks, and how we control them



Citizen data

- Office owns the data; we only process it
- Hosted in India; keys held by the office
- Encrypted, OTP login, full audit log
- Masked before any AI call (detail: previous slide)



AI mistakes

- AI suggests, an officer decides
- Under 70% confidence → a person checks
- Fixed department list; AI cannot invent one
- Accuracy and bias reviewed monthly



System and vendor

- AI down → routing falls back to the clerk
- Daily backups; works on slow connections
- CERT-In empanelled security audit
- Open source; data exportable any time



Staff adoption

- Hindi first, fewer steps than the register
- A champion clerk in each department
- 30-min desk-side training; paper kept 4 weeks
- DM reviews from the dashboard in meetings

E. The pitch, the objections, and the price

> In ten weeks, in two departments, every application gets a tracking number and reaches the right desk on day one, and you see every delayed file on one screen. Your staff still decide. Data stays in India. If delays don't fall by a third, you don't pay for the next phase.

"We already have a system." We fill gaps around CPGRAMS and e-Office, and feed data back.

"It is not secure." India hosting, OTP login, audit log, security audit.

"Staff won't use it." Hindi, fewer steps, champions; usage tracked weekly.

Pilot · 10–12 weeks

₹5–8 lakh

Fixed fee, 2 departments; next phase only if targets are met

Rollout · Module by module

₹20–30 lakh

Whole district, bought through GeM

Support · Yearly

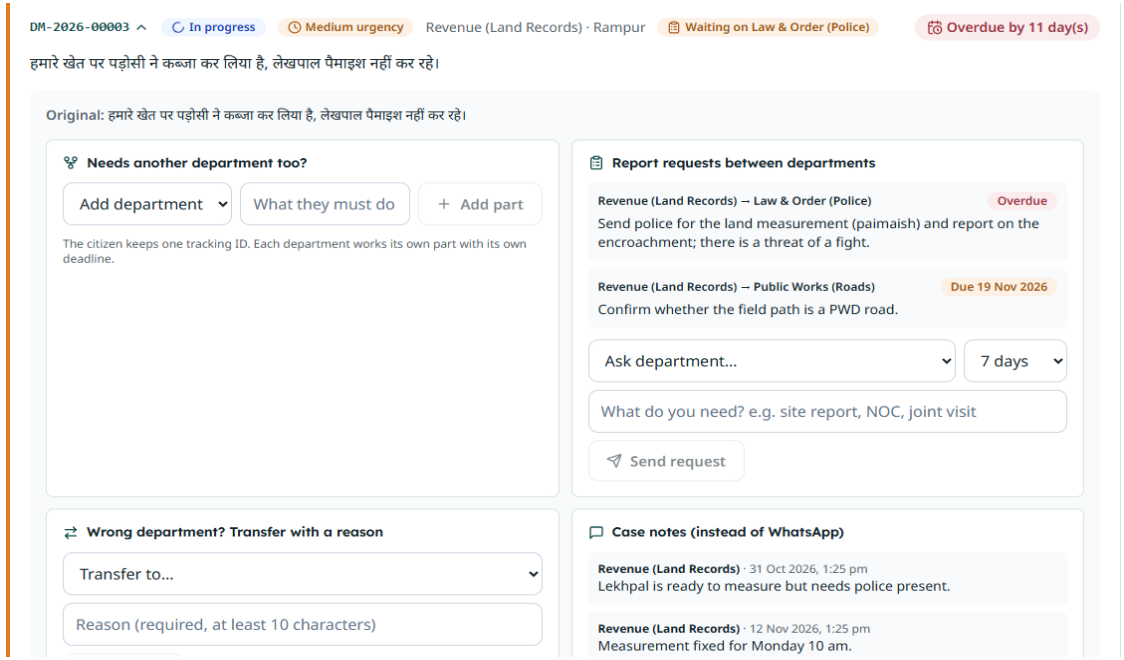
15–20% / yr

Of build cost: hosting, helpdesk, AI monitoring, training

Indicative figures, firmed up after discovery. Discovery can be free, to earn trust.

F. Working proof: AI grievance router, live

- Complaint in Hindi, English or a mix; phone and Aadhaar masked before the AI sees it
- AI suggests department, urgency and a summary, with a confidence score
- Under 70% confidence → clerk review queue; every decision logged
- Several problems in one complaint → joint case: one tracking ID, one part per department
- Departments talk inside the case: transfer with a reason, report requests with deadlines, notes, inbox, DM circulars
- Second transfer needs ADM approval; dashboard shows who is holding others up
- 22 test cases incl. 3 joint ones: Claude Sonnet 100% correct, 0 wrongly routed



Live: smart-collectorate-grievance-router.vercel.app

Code: github.com/easycodewithme/Assignemnt-Automation

Next steps: 1. Approve a 3-week discovery (no cost) 2. Name a nodal officer and two pilot departments 3. Agree targets and the go / no-go date

Appendix: 50 questions by person and department (1 of 2)

DM / ADM · Collector's office

1. Which three numbers do you want on your desk every Monday?
2. Which departments' delays reach you most often, and why?
3. Who may transfer, reject or close a grievance? Who approves a second transfer?
4. How are joint problems (sewer + water + power) handled today, and who leads?
5. Which two departments and which tehsil should go first?
6. What would make you stop the pilot, and what counts as success?

Grievance clerk · Inward desk, Collectorate

7. Walk us through the register: how is a complaint numbered and marked?
8. How many come in a day by counter, post, CPGRAMS / IGRS and phone?
9. Which complaints are hardest to mark to one department?
10. How often does a file come back as “not our department”?
11. How much of your day goes on “what is my status?” visits and calls?
12. How long does the weekly review list take to prepare?

Tehsildar / SDM · Revenue

13. Which land cases need Police or PWD before you can act?
14. How do you ask another department for a report today, and how long does it take?
15. Where does a mutation or measurement file wait longest?
16. Which land-record system (Bhulekh) must we read from?

Lekhpal / Patwari · Revenue, field

17. How do field verification tasks reach you, and how many a week?
18. How do you send the report back: paper, WhatsApp or an app?
19. Do you have a smartphone and network in the villages you cover?

Executive Engineer · Electricity (DISCOM)

20. Should we link to your 1912 helpline or app, or forward cases to it?
21. Who decides if it is a feeder, transformer or billing problem?
22. What is your target time for a burnt transformer and a wrong bill?

Executive Engineer · Jal Nigam / PHED

23. Who owns hand pumps, Jal Jeevan taps and town supply?
24. When supply fails because of a sewer, power cut or road digging, who coordinates?
25. What must the citizen tell us so a team can go (ward, pump or pipeline ID)?

Appendix: 50 questions by person and department (2 of 2)

Executive Officer · Municipal Council

26. Which areas are under the municipality and which under the panchayat?
27. How do you hear about a sewer overflow today, and who is sent?
28. Do you already use a grievance app we should link to instead of copying?

Executive Engineer · PWD

29. How do we tell a PWD road from a panchayat or municipal road?

District Supply Officer · Food & Civil Supplies

30. Which ration complaints can be checked in the ePoS data without a visit?
31. Who inquires against a ration shop, and how long does it take?

BDO / PD-DRDA · Rural Development

32. Which portal shows MGNREGA wage and PM Awas payment status?
33. Which cases need Revenue to confirm land first?

District Social Welfare Officer · Social Welfare

34. Which pension complaints are bank / PFMS failures, and which are pending approval?
35. Can a clerk check pension payment status without your office?

CMO · Health

36. Which complaints must reach you within hours (ambulance, Ayushman refused)?
37. Who answers for staff absence at a PHC or CHC?

SP office grievance cell · Police

38. Which complaints go straight to the police station, and which to the SP office?
39. What police information must never be shown to other departments?

District Informatics Officer · NIC

40. Where must it be hosted: NIC cloud, State Data Centre or empanelled cloud?
41. Can we connect to CPGRAMS, IGRS / Jansunwai and e-Office? How?
42. May masked text go to an AI service, or must AI run in India?
43. Which security audit and approvals are needed before go-live?
44. How do staff log in today (Parichay / single sign-on)?
45. What are the rules for record retention, RTI and breach reporting?

Citizens · At the gate and on the helpline

46. How many visits did this complaint take, and what did each cost you?
47. Did you have to go to more than one office for one problem?
48. Would you rather write, speak, or have a clerk enter your complaint?
49. How do you want updates: SMS, WhatsApp or a call?
50. Were you told the outcome, and was it really fixed?